PROJECT REPORT

🏷️ **Title: Retail Business Performance & Profitability Analysis**

1️⃣ Introduction

In today's competitive market, businesses need to understand their sales and profitability to make better decisions. This project focuses on analysing retail data to identify key trends in sales, profit, and customer behaviour.

The main objective is to find out:

- Which products generate profit
- Which regions perform best
- Where the business is facing losses

2️⃣ Objectives

The objectives of this project are:

- To analyse sales and profit across different product categories
- To identify high-performing and low-performing regions
- To detect loss-making products and categories
- To create interactive dashboards for business insights

3️⃣ Dataset Description

The dataset used in this project is the **Superstore dataset**, which contains retail transaction data.

📊 **Key Columns:**

- Order ID
- Order Date
- Ship Date
- Customer Name
- Region
- Category (Furniture, Technology, Office Supplies)
- Sub-Category
- Sales
- Quantity

- Discount
- Profit

4️⃣ Tools & Technologies Used

- **Python (Pandas, Matplotlib, Seaborn)** – Data cleaning and analysis
- **SQL** – Querying and aggregation
- **Power BI / Tableau** – Data visualization and dashboard creation

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5️⃣ Data Cleaning & Preprocessing

Before analysis, the dataset was cleaned:

- Removed missing/null values
- Checked for duplicate records
- Converted date columns into proper format
- Ensured correct data types for numerical columns

6️⃣ Data Analysis

📌 1. Profit by Category

- Technology category generated the highest profit
- Furniture category showed losses in several cases

📌 2. Sales by Region

- West region recorded the highest sales
- Central region showed lower performance

📌 3. Top Performing Products

- Some products contributed significantly to profit
- Identified top 10 products based on sales and profit

📌 4. Loss-Making Products

- Certain furniture items had negative profit
- High discounts were a major reason for losses

📌 5. Impact of Discount on Profit

- Higher discounts led to reduced profitability
- Optimal discount range identified for better profit

7️⃣ Data Visualization

Visualizations created include:

- Bar chart: Profit by Category
- Pie chart: Sales distribution by Region
- Line chart: Sales trend over time
- Bar chart: Top 10 products
- Scatter plot: Discount vs Profit

These visualizations help in understanding patterns easily.

8️⃣ Dashboard Overview

An interactive dashboard was created using Power BI/Tableau with:

- KPI metrics (Total Sales, Total Profit)
- Category-wise performance
- Region-wise analysis
- Monthly sales trends

9️⃣ Key Insights

- Technology products are the most profitable
- Furniture category needs improvement due to losses
- Excessive discounts reduce profit margins
- West region performs better compared to others
- Identifying loss-making products helps in decision-making

🔟 Conclusion

This project successfully analyzed retail business data to identify key performance areas. By understanding sales trends, profit distribution, and customer behavior, businesses can make better strategic decisions.

The insights gained from this analysis can help improve profitability and optimize business operations.

1️⃣ Future Scope

- Implement predictive models for sales forecasting
- Customer segmentation for targeted marketing
- Real-time dashboard integration
- Advanced analytics using machine learning